# Notes_25.10: kNN Classification Code in Python










🌐 **Language: English** | हिंदी / Hinglish | ⬅️ Lecture 25 Overview

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## 📌 Executive Summary

Implementing KNeighborsClassifier on heart disease data with feature scaling.

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## 🎯 Mathematical Formulations

Uniform vs Distance-Weighted Voting:

$$ w_i = \frac{1}{d(\mathbf{x}, \mathbf{x}*i)}, \qquad \hat{y} = \arg\max_c \sum{i \in N_K, y_i = c} w_i $$

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## 💡 Conceptual Principles & Architecture

### Feature Scaling kNN Ke Liye Compulsory Hai!

kNN geometry aur distances par nirbhar karta hai. Agar hum data scale nahi karenge, toh bada feature distance ko dominate karega aur accuracy dramatically drop ho jayegi.

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# 💻 Python Production Implementation

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score

df = pd.read_csv("heart.csv")
X = df.drop("target", axis=1)
y = df["target"]

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_s

scaler = StandardScaler()
X_train_s = scaler.fit_transform(X_train)
X_test_s = scaler.transform(X_test)

knn = KNeighborsClassifier(n_neighbors=5, weights='distance')
knn.fit(X_train_s, y_train)
print("kNN Accuracy:", accuracy_score(y_test, knn.predict(X_test_s)).round(4))
```